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July 27, 1972

SUPERSEDING
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FEDERAL SPECIFICATION

SEALER, SURFACE RENEWER FOR WOOD AND SEALER FOR CONCRETE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a material for use on wood floors as a sealer and surface renewer and concrete floors as a sealer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

P-D-220 - Detergent, General Purpose, Solvent Type.
P-W-155 - Wax, General Purpose, Solvent Type.
TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
TT-S-735 - Standard Test Fluids; Hydrocarbon.
TT-T-775 - Tung oil (China-wood) for use in organic coating.

Federal Standard:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling and Testing.

Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, Mo, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 Material. The sealer as received shall be a good commercial quality entirely suitable for the purpose intended and shall consist of tung oil conforming to TT-T-775 blended with phenolic resin modified with rosin and pentaerythritol. Plasticizers and other additives including drier may be used provided that the finished product meets all the requirements of this specification. The solvent or thinner used in the process shall comply "Rule 66"[1] (air pollution regulations). A certificate of compliance to this effect is necessary.

[1] Information on Rule 66 may be obtained from Air Pollution Control District Cound of Los Angeles, Los Angeles, California 90013.

FSC 8010

3.2 Qualitative requirements.

3.2.1 Condition in container. The material as received shall show no curdling, gelling, skinning or putrefaction when tested as in 4.3.1.

3.2.2 Appearance. When examined as in 4.3.2 the material as received shall be clear, transparent, and homogeneous. The dried film, when tested as in 4.3.2 shall be smooth, even, and shall show no orange peeling or other film irregularities.

3.2.3 Storage stability.

3.2.3.1 Partially full container. The material, when tested as in 4.3.3.1 shall not skin within 48 hours.

3.2.3.2 Full container. When stored for six months as tested in 4.3.3.2, shall not show livering, skinning, and a change of viscosity as specified in table I.

TABLE I. Quantitative requirements

Characteristics	Requirements Minimum	 Maximum
Nonvolatile, percent by weight of sealer	28	--
Color (Gardner color standards)	--	9
Viscosity (Gardner tubes)	--	A4
Flash point (degree F.)	103	--
Drying time:		
Set to touch (hours)	1-1/2	2-3/4
Dry hard (hours)	4	8
Abrasion resistance, wear cycles	200	--
Slipperiness-coefficient of friction (James Machine)	0.5	--

3.2.4 Odor. The odor in the can, during application, and after drying shall not be irritating or offensive when tested as in 4.3.4.

3.2.5 Application properties. The sealer when applied by brush or lamb's wool mop on wood panel shall show uniform penetration and shall not obscure the natural color of the wood when tested as in 4.3.5. The second coat shall not damage the first coat.

3.2.6 Flexibility. When tested as in 4.3.6 the dried film of the sealer shall withstand bending without cracking or flaking from the panel.

3.2.7 Resistance tests.

3.2.7.1 Resistance to cleaning solutions. When tested as in 4.3.7.1, the film shall show no dulling, whitening, blistering or flaking when examined half hour after removal.

3.2.7.2 Resistance to hydrocarbon. A film of the material prepared and tested as in 4.3.7.2 shall show no blistering, whitening, or wrinkling when examined half hour after removal.

3.2.7.3 Resistance to water. When tested as in 4.3.7.3, the film shall show no blistering, wrinkling, or whitening when examined half hour after removal.

3.2.7.4 Resistance to wax solution. When tested as in 4.3.7.4, the film shall show no blistering, wrinkling, or whitening when examined half hour after removal.

3.3 Quantitative requirements.

3.3.1 The quantitative requirements shall be as specified in table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Sampling and inspection. Sampling and inspection shall be performed in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Production inspection of the sealer.
(b) Inspection of preparation for delivery.

4.3 Test procedures. Test procedures shall be conducted in accordance with Fed. Test Method Std. No. 141 except as otherwise specified in table II.

TABLE II. Index

Characteristics	Requirement reference	Applicable Tests: Fed. Test Method Std. No. 141	Applicable Tests: Paragraph reference
Condition in container	3.2.1	----	4.3.1
Appearance	3.2.2	----	4.3.2
Storage stability, partially full container	3.2.3.1	----	4.3.3.1
Storage stability, full container	3.2.3.2	----	4.3.3.2
Odor	3.2.4	----	4.3.4
Resistance to cleaning solution	3.2.7.1	----	4.3.7.1
Resistance to hydrocarbon	3.2.7.2	----	4.3.7.2
Resistance to water	3.2.7.3	----	4.3.7.3
Resistance to wax solution	3.2.7.4	----	4.3.7.4
Nonvolatile matter	Table I	4041	---
Color	Table I	4248	---
Viscosity	Table I	4271	---
Flash point	Table I	4291	---
Drying time	Table I	4061	---
Abrasion resistance	Table I	6192	---
Slipperiness	Table I	----	4.3.8

4.3.1 Condition in container. Determine packaged condition of the sealer in accordance with method 3011 of Fed. Test Method Std. No. 141 for compliance with 3.2.1.

4.3.2 Appearance. Transfer a portion of the packaged sealer into a clean, dry test tube, examine by transmitted light and evaluate for compliance with 3.2.2. Spread (by brush or cheesecloth) a portion of the sealer on a clean glass panel and allow to dry for 24 hours at room temperature. Then examine for compliance with 3.2.2.

4.3.3 Storage stability.

4.3.3.1 Partially full container. Determine the 48 hour skinning in accordance with method 3021 of Fed. Test Method Std. No. 141 for compliance with 3.2.3.1.

4.3.3.2 Full container. In accordance with method 3022 of Fed. Tes Method Std. No. 141, allow a full quart can of sealer to stand undisturbed for six months and then examine the content for compliance with 3.2.3.2. A certificate of compliance from the supplier is acceptable.

4.3.4 Odor. Observe the odor in the can, during application, and after the sealer has dried 24 hours. Then evaluate for compliance with 3.2.4.

4.3.5 Application properties. Prepare a 2-1/4 x 12 inch panel of First Grade Maple Flooring in the following manner: Sand the surface of the wood with 00 sandpaper until flat and smooth, and remove excess sanding dust with a clean, dry cloth. Apply a coat of the sealer to a dry film thickness of about 1 mil and let dry for 24 hours at a temperature between 70° to 90°F. Examine for compliance with 3.2.5. Apply a second coat at the same thickness and allow to dry for 48 hours. Cut a series of 8 parallel cuts using sharp razor blades. The cuts shall be 1/8 inch apart. A second series of similarly spaced cuts shall be made at right angles to the first. Examine the cuttings for penetration and compliance with 3.2.5. Evaluate the second coat for compliance with 3.2.5.

4.3.6 Flexibility. Flow a coat of the sealer on a tin panel prepared in accordance with method 2012 of Fed. Test Method Std. No. 141 and let stand in a vertical position 75° ± 5°F., 50 ± 5 percent relative humidity for 1 hour. The thickness of the dried film shall be between 1 to 1-1/2 mils. Place in horizontal position for 48 hours under the same conditions. Then place the panel in a well ventilated oven at 160°F. for 4 hours. Remove and allow to cool at room temperature for 1 hour. Bend over 1/8 inch mandrel in accordance with method 6221 of Fed. Test Method Std. No. 141 and observe for compliance with 3.2.6.

4.3.7 Resistance tests: (Prepare four glass panels (method 2021) by brushing a coat of sealer to a minimum dry film thickness of 1 mil and allow to dry for 24 hours).

4.3.7.1 Resistance to cleaning solution. Make a five percent solution of a heavy duty detergent conforming to P-D-225. Immerse one of the panels prepared in 4.3.7 in the detergent solution for 2 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. Remove the panel and allow to dry for 1/2 hour and evaluate for compliance with 3.2.7.1.

4.3.7.2 Resistance to hydrocarbon. Immerse one panel (prepared in 4.3.7) in the test fluid conforming to TT-S-735 1/ for two hours in accordance with method 6011 of Fed. Test Method Std. No. 141. Remove the panel, allow to dry for 1/2 hour and evaluate for compliance with 3.2.7.2.

4.3.7.3 Resistance to water. Immerse one panel (prepared in 4.3.7) in distilled water for 2 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. Remove the panel, allow to dry for 1/2 hour, and evaluate for compliance with 3.2.7.3.

4.3.7.4 Resistance to wax solution. Immerse the last panel (prepared in 4.3.7) in wax solution conforming to P-W-158 for two hours in accordance with method 6011 of Fed. Test Method Std. No. 141. Remove the panel, allow to dry for 1/2 hour, and evaluate for compliance with 3.2.7.4.

4.3.8 Slipperiness (coefficient of friction). This test shall be run on James Machine 2/. This method of test covers a procedure for the measurement of the static coefficient of friction of floor finishes. A certificate of compliance from the supplier is acceptable.

4.3.9 Abrasion resistance. Determine abrasion resistance of the sealer in accordance with method 6192 of Fed. Test Method Std. No. 141 using 250 grams load and wheel No. CS-17. Apply the sealer on the test plate (see 2.1.6 of the method) by brush to a minimum 1 mil dry film thickness and allow to dry for 48 hours at room temperature. Run the test and determine the wear cycles for compliance with table I.

4.3.10 Packaging, packing, and marking. The sealer shall be inspected for compliance with the packaging, packing, and marking requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The sealer shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be level A, B, or C, and the level of packing shall be level A, B, or C as specified (see 6.2). The sealer shall be furnished in 1-gallon metal cans, 5-gallon steel pails, and 55-gallon metal drums as specified (see 6.2).

5.2 Special marking.

5.2.1 Direction for use. The directions for use, which shall be clearly legible, shall be shown on the reverse side of the container and shall read as shown in 5.3.

5.3 DIRECTIONS

Use for sealing new interior wood floors or for recoating varnished wood floors. Also, use for sealing concrete walls and floors where there is no hydrostatic pressure. Concrete must be completely cured before coating.

1/ Type III

2/ This machine was developed by S.V. James of the Underwriters' Laboratories, Inc., 207 E. Ohio Street, Chicago, Illinois. The test procedure is included in the operational description of the machine.

SURFACE PREPARATION AND APPLICATION

Wood floors. Open grained new wood floors such as oak should be filled with a paste wood filler. Old floors should be cleaned of all wax and oil with an alkali cleaner. Rinse and let dry. Sand lightly to remove any gloss remaining from old finish. All floors must be clean and dry.

Use either lamb's wool or felt applicators, brushes or soft cloths. Apply freely and evenly, leaving no excess on surface. Apply 3 coats on new wood, 1 coat on finished wood.

Concrete floors. Remove all dirt, grease and oil. Fill cracks with patching cement. If concrete is smooth and slick, roughen with acid type floor etcher.

Use clean, dry brushes or rollers. Apply 2 coats on bare concrete, 1 coat on previously sealed concrete.

Thinners. Do not thin. Apply sealer-surface renewer just as it comes from the container.

6. NOTES

6.1 Intended use. This sealer-renewer is intended for cleaning and renewing all hardwood floors and concrete, particularly in heavy traffic areas. It helps preserve natural wood beauty. This material renews areas where finish is worn. It helps replace seal lost through wear. This sealer prevents concrete floors from dusting and minimized spalling.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a) Title, number, and date of this specification.
b) Size of container required (see section 5).
c) Levels of packaging and packing (see section 5).

CIVIL AGENCY INTERESTS:

GSA-FSS
PO-OF
VA-DMS
HEW-NIH
DOT-CGS
INT-[illegible]

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.